

here is a very simple **linear transformation** example.

Example: Scaling

Suppose we have

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

This means

$$x' = 2x, \quad y' = 3y$$

So the point

$$(1, 2)$$

becomes

$$(2, 6).$$

What does the matrix do?

$$\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

means:

- x is multiplied by 2 → horizontal stretching
- y is multiplied by 3 → vertical stretching
- There is **no translation**

So this is a **linear transformation**.

Important distinction for your students

A transformation is **linear** if it has the form

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = A \begin{bmatrix} x \\ y \end{bmatrix}$$

and therefore **the origin must remain fixed**:

$$(0,0) \rightarrow (0,0).$$

For example,

$$x' = 2x + 2$$

is **not** a linear transformation because of the $+2$. That's an **affine transformation**.

Simple hierarchy:

$$\text{Linear} \subset \text{Affine} \subset \text{Homography}$$

where each step allows more types of transformations.